Knee MRI Abnormality Detection with LLM-Read Report Labels: Approaches and Validation Lessons from the RSNA Kaggle Challenge

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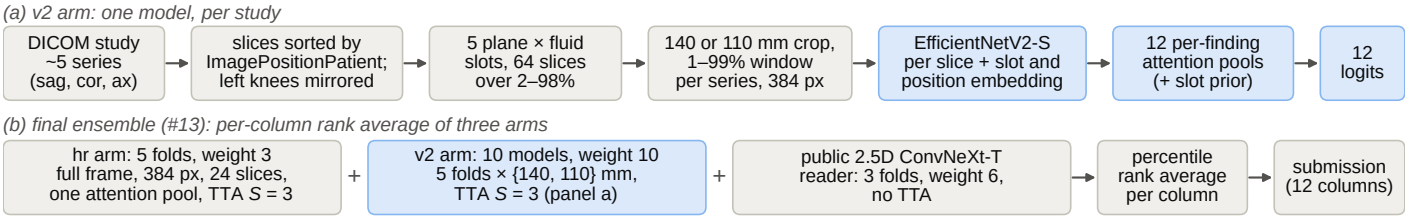


Figure 1: Pipeline. (a) One v2 model: anatomically ordered, laterality-canonical slices in five plane × fluid slots; a shared EfficientNetV2-S; 12 per-finding attention pools. (b) Submission: per-column percentile-rank average of three arms (3 : 10 : 6).

Abstract

We describe our entry to the RSNA Knee Abnormality Detection challenge on Kaggle: 12 findings per knee MRI study, scored by macro ROC AUC, with radiologist labels for only 58 of 4,407 training studies. We trained on labels that large language models (LLMs) extracted from the reports and rank-averaged 15 EfficientNetV2-S models with a public 2.5D ConvNeXt-T reader, scoring **0.927 private** (0.941 public), rank 3,011 of 6,299: a mid-table result against a top private score of 0.960. Model choices were gated on 5-fold out-of-fold (OOF) AUC and the 58 gold studies. Private ran 0.014–0.020 below public with our 15 submissions in the same order, and our first-choice final was our best private entry; the gap to the top came from the approach, not from overfitting. Local gains transferred for our own models trained on report labels, but not for distillation on OOF-derived targets or for a validated public model (+0.0069 OOF; +0.001 public, +0.002 private).

1 Task and data

The challenge [1] asks for a probability of each of 12 findings per study (ACL, MCL, medial and lateral meniscal tear, medial, lateral and patellofemoral osteoarthritis, effusion, synovitis, Baker's cyst, contusion, fracture), scored by the mean of the 12 ROC AUCs, so only within-column ranks matter. The 4,407 training studies (about five series each) have free-text reports in several languages but only 58 radiologist labels. The ≈1,300 hidden test studies have images only and are predicted offline on one T4 GPU within 9 h; public and private scores use disjoint subsets.

2 Labels from reports

Targets come from public LLM-extracted report labels [2]: first one soft table (stevenleehans v4; 0.893 AUC on gold), finally a soft consensus of three (stevenleehans v2, pilkwang v2, lixin73) with per-cell weight $w = 0.15 + 0.85\,(0.65\,a + 0.35\,c)$ from teacher agreement a and certainty c, plus the gold studies at $w = 3$, as in the public notebooks. Every table was leak-checked against gold first (one public merge reproduces all 58 labels). A local Gemma-3-27B [9] re-read for synovitis lost on gold (0.722 vs. 0.790): 14 of its 27 gold positives have reports that never mention the synovium.

3 Models and ensemble

v2 arm (Fig. 1a). Following pilkwang's public baseline and its descendants, each chosen series is decoded once, centre-cropped to 140 or 110 mm (from `PixelSpacing`), windowed at its own 1st–99th percentiles and resized to 384 px. Slices are ordered by `ImagePositionPatient` (file order matched anatomy in none of 400 sampled series) and fill five plane × fluid slots (sagittal 18 + 14, coronal 12 + 8, axial 12) spaced over 2–98% of the stack. Left knees are mirrored to a right-knee frame (side from the DICOM tag or, when absent, image-centre geometry; 97.5% agreement). An EfficientNetV2-S [4] (timm [5]) embeds each slice; with slot and position embeddings, 12 per-finding gated-attention pools [6] with a fixed anatomical slot prior (+0.55 logit) feed 12 linear heads. Training: weighted soft binary cross-entropy, 38 slices per study (64 at test), AdamW at 3×10^{-4} for all layers, cosine schedule, EMA, 10 epochs (4.3–4.8 h per fold on an NVIDIA GB10). Our first v2 run, with the 0.1× backbone rate (2×10^{-5}) of the public ViT notebooks, scored 0.031 lower on fold 0.

hr arm. An older high-resolution, full-frame arm (24 slices in file order, i.e. at effectively random positions; per-slice min–max; one attention pool; stevenleehans v4 labels) is weak alone (0.871 on fold 0) but different.

Ensemble (Fig. 1b). Per-column percentile ranks are averaged with weights hr 3, v2 10 (five 140 mm and five 110 mm fold models) and reader 6 (Sec. 6), set on a coarse grid on stitched OOF; hr and v2 models also run on S=3 phase-shifted slice grids (slot-phase test-time augmentation, TTA). Series are decoded once per crop; the hidden run takes about 5 h.

4 Validation protocol

Each candidate gets (i) stitched 5-fold OOF macro AUC on all 4,407 studies against the binarised stevenleehans v4 labels, each study scored by the model that did not train on it, and (ii) AUC on the 58 gold studies, each scored by the fold that held it out: one large but report-based, one true but with 9–35 positives per finding. Gates were fixed in advance (e.g. a new member had to add ≥ +0.003 OOF without lowering gold). Submissions needed about +0.005 expected gain (the board shows three decimals) or served as a pre-registered calibration of a new kind of member. Finals were picked by local validation under a decision table fixed in advance; nothing was fitted to the board.

5 Results

Figure 2 shows all submissions. The big steps were fold ensembling of the 224 px baseline (+0.016 public, +0.020 private; invisible to OOF, where each study is scored by one

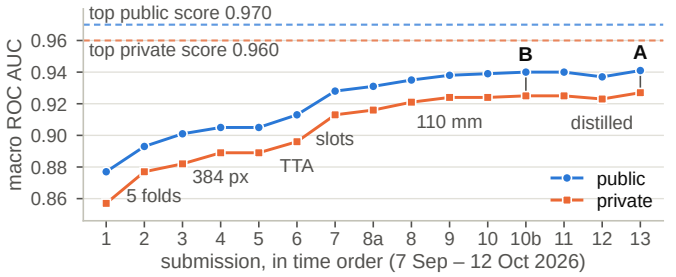


Figure 2: All 15 submissions in time order (numbered as in our log). Notes name the change behind a step; A, B: finals.

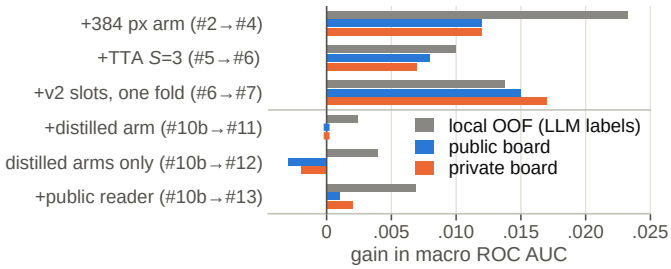


Figure 3: Local vs. board gain of single changes (OOF on LLM labels; v2 row: fold 0 only; reader: 2,668 studies).

model), the 384 px arm, TTA and above all the slot contract: one fold-0 v2 model added +0.015/+0.017 and the full v2 arm reached 0.940/0.925 (#10b). Private ran 0.014–0.020 below public and no two submissions swapped order (ties aside; Kendall $\tau_b = 0.995$). Final A (#13, best local validation) is also our best private score; B is #10b (own models only). We rose from 3,163rd (public) to 3,011th (private). For our own v2-era entries (#8–#10b), public−OOF stayed at +0.048 to +0.050 and private−OOF at +0.034 to +0.035.

In Figure 3, the first three changes transferred at 0.5–1.2× their OOF gain; the others did not. *Distillation*, retraining v2 arms on targets mixed 50/50 from the consensus and the v2 arms' own OOF predictions, raised OOF by up to +0.0040 (gold +0.0074) yet scored 0.000/−0.003 public and 0.000/−0.002 private: a student's targets carry information from models that saw its held-out fold, so both local anchors were optimistic.

6 Validating a public model

goodpjw2008's reader [3] (Apache-2.0; ConvNeXt-T [7] on slice triplets, a study-level transformer, per-finding pools) has a recoverable split: its GroupKFold over report text rederives from the competition CSV and excludes gold. We ported its inference verbatim and proved the split: each checkpoint is 0.057–0.067 worse on its own re-derived fold than on its training folds, while a wrong split shows no gap. Its honest OOF on 2,610 held-out studies is 0.8885 (gold 0.9105). At weight 6 (chosen on OOF from a pre-registered grid) it adds +0.0069 OOF (95% CI +0.0058 to +0.0080) and +0.0023 gold; a leaky three-checkpoint mean would have claimed +0.0235. The board gave +0.001 public and +0.002 private (about 1/7 and 2/7 of that), though its Kaggle outputs matched ours to 3×10^{-4}: a recovered split is necessary, not sufficient.

7 Engineering lessons

Four silent failures that would each have shipped an all-0.5 file were caught before submitting: a P100 that Kaggle's PyTorch no longer supports (we pin the T4), bf16 convolutions on the T4 (fp16 instead), a JPEG-trained pipeline meeting the DICOM test mount, and a fingerprint check aborting on cross-GPU drift. The costliest bug was quiet: the 110 mm checkpoints stored the default 140 mm crop and the notebook used the first member's crop for all, so in two submissions the 110 mm arm saw 140 mm crops (−0.0106 on fold 0); reading the kernel log revealed it. The 9 h limit capped the ensemble: 20–25 v2 models at S=3 would need 8.3–9 h.

Idea (closed)	ΔAUC	Measured on
512 px at the 110 mm crop	−0.0031	fold-0 model
2.5D context ($j \pm 1$ as channels)	0.0000	fold-0 model
170 mm crop as a 4th arm	+0.0004	fold-0 ensemble
CoAtNet-2, public recipe	+0.0003	fold-0 ensemble
Second seed of both v2 arms	+0.0005	final ensemble
TTA S=5 instead of 3	+0.0001	stitched ensemble
Cross-crop TTA	$\leq$ +0.0002	fold-0 ensemble
Per-column arm weights	+0.0004	nested 5-fold
Protocol features (series counts)	+0.0002	stitched ensemble
Distillation (Fig. 3)	+0.0040	stitched ensemble

Table 1: Closed ideas (Δ OOF AUC on LLM labels); CoAtNet-2 and protocol features also cost gold (−0.0017, −0.0013).

8 Limits and next steps

Table 1 lists closed ideas: with the slot contract in place the ensemble saturated near 0.890 OOF, and CoAtNet-2 [8], trained with dreaddevelopment's public "Raptor" recipe, matched our models alone (0.878 on fold 0) but made the same errors. Ten days before the deadline the 0.95x public notebooks rested on two public models: nartaa's CoAtNet-2 (0.949 alone), trained on all training studies plus 2,399 external OAI knees, and heliosli's ConvNeXt-S, trained on LLM-graded report "states" with segmentation-anchored 100 mm local views (0.954 blended); per-finding blend constants added ≤ 0.001 by their authors' accounts. We used neither (no held-out predictions for the first; the second not pursued after the reader's weak transfer). The gap to the 0.960 private top thus points to data, labels and views, not to architecture or overfitting. Next time we would fix the input contract and graded labels first, spend GPU-hours on data and views rather than crop and backbone sweeps, and keep the validation discipline that left the private board without surprises.

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